ESE 123 Lab 1 Report: I-V Characteristics of a Resistor and an LED

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Objective. Use the power supply and DMM to measure current vs. voltage for a 1 Ω resistor and an LED, find the resistance from the slope, and see why an LED does not follow Ohm's law.

1. Measured data

1 Ω resistor			**LED**		
Set V	Measured V	Measured I	Set V	Measured V	Measured I
0	0.000179 V	0.000199 A	0	−0.001 V	−0.00015 A
0.05	0.03556 V	0.036 A	0.5	0.500 V	0.000004 A
0.10	0.0709 V	0.071 A	1.0	1.001 V	0.000003 A
0.15	0.105 V	0.107 A	1.5	1.501 V	0.000003 A
0.20	0.1398 V	0.141 A	2.0	2.001 V	0.000002 A
0.25	0.175 V	0.176 A	2.5	2.501 V	0.00088 A
0.30	0.211 V	0.212 A	3.1	2.855 V	0.00529 A
0.35	0.2465 V	0.248 A	3.1	3.022 V	0.010322 A
0.40	0.2819 V	0.283 A	3.1	3.100 V	0.012900 A
0.45	0.317 V	0.319 A	3.1	3.100 V	0.012977 A
0.50	0.351 V	0.352 A			

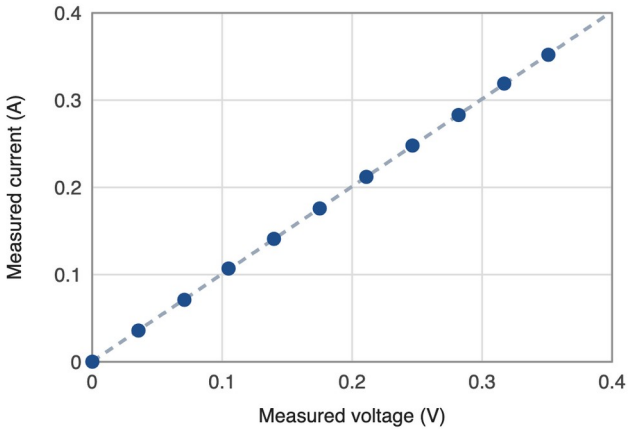


Figure 1: Resistor. Dashed line: fit I = 1.003·V + 0.00045 (R^2 = 1)

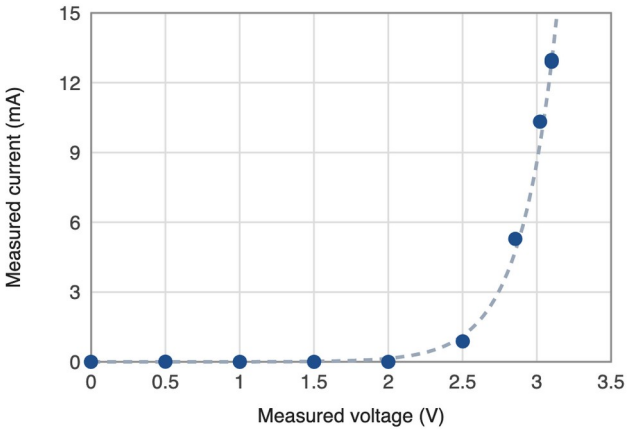


Figure 2: LED. Dashed line: fit I = 4.15×10^{-8}·e^(4.08·V) (R^2 = 0.998)

2. Resistor (nominal 1 Ω, ±5%)

> Q1: G = ΔI/ΔV = slope of the fit = 1.003 S
> Q2: R = 1/G = 1/1.003 = 0.997 Ω
> Q3: error = (0.997 − 1)/1 × 100% = −0.3 %

Q4: Yes, 0.997 Ω is inside the 0.95 to 1.05 Ω range. Possible error sources: resistance of the wires and contacts ($R_{measured} = R_{resistor} + R_{wire} + R_{contact}$), DMM and supply accuracy, and the resistor warming up at about 0.35 A.

3. LED

Q5: A straight-line fit through all of the LED points gives a slope of 3.86 mS, or about **259 Ω**. But the line fits badly. The LED passes almost no current up to 2 V, then the current rises sharply above about 2.5 V. Its I-V curve is exponential, not a straight line, so the LED does **not** follow Ohm's law and has no single resistance.

Q6 and Q7: Differential resistance $R_d = \Delta V/\Delta I$ between two neighboring points:

> High current: $R_d = (3.100 - 3.022) / (0.012900 - 0.010322) = 0.078 / 0.002578 \approx 30\ \Omega$
> Low current: $R_d = (2.501 - 2.001) / (0.00088 - 0.000002) = 0.500 / 0.000878 \approx 570\ \Omega$

At low current the LED behaves like a large resistance. Once it turns on (and glows brighter), a small increase in voltage gives a large increase in current, so R_d drops to about 30 Ω. Below 2 V the current is only a few µA (near the DMM's resolution), so R_d there is effectively very large.